

Pharmacology of treatment of Gastritis and Peptic ulcer

A- Anti-secretory drugs

1. Proton Pump Inhibitors (PPIs)

Omeprazole - Esomeprazole – Lansoprazole - Dexalansoprazole - Pantoprazole - Rabeprazole (oral – IV - 20-40 mg/d).

Mechanism of Action

Act on the final step of HCl secretion → Most effective acid suppressants.

They are prodrugs → after absorption → parietal cells → secretory canaliculi → converted by acid into active form → inhibits H⁺/K⁺ ATPase (proton pump) → Inhibit basal & meal stimulated acid secretion (98%, 1-2 h after 1st dose).

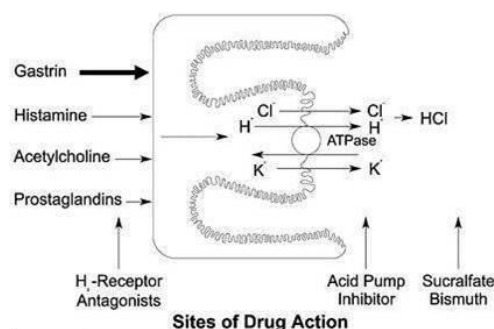
They have anti- H. pylori effect (Lansoprazole most potent).

Pharmacokinetics

1. Weak bases → destroyed by gastric acidity & ionized in acid stomach (not readily absorbed) → given in enteric-coated form.
2. Should be taken 30 – 60 minutes before meals (on empty stomach): only inhibit active pumps - Food → ↓ bioavailability by 50%.
3. Repeated administration → ↓ gastric acidity → ↑ bioavailability.
4. Long duration (onset 1 hour & duration 24 hours (t_{1/2} = 1 h but PPIs covalently bind to PP and synthesis of new pump requires 24 hours).
5. Metabolized in the liver.

Adverse effects

1. GIT: abdominal pain, nausea, vomiting, diarrhea or constipation.
2. CNS: headache, dizziness, somnolence.
3. Hypochlorhydria → ↑ Risk of infection (e.g. hospital acquired pneumonia & H. pylori) and Gastric tumor (in animals): bacterial colonization → ↑ carcinogenic nitroso compounds.
4. Vitamin B12 deficiency (on long term use)
5. Enzyme inhibition (clinically insignificant but inhibit activation of clopidogrel).



2. Potassium Competitive Acid Blockers (P-CAB)

Vonoprazan, revaprazan, tegoprazan (newer version of PPIs): differs from PPIs in:

Omeprazole	Vonoprazan
Irreversible PP inhibition (↓ active pumps)	Competitive reversible PP inhibition (↓ both active and inactive PP)
Takes 3-5 days for maximal acid suppression	Take 1 day for maximal acid suppression
T _{1/2} = 1-2 hrs (but longer duration of action being irreversible)	T _{1/2} = 6-9 hrs ³³
Prodrug	Not Prodrug
Not acid stable (Taken 30-60 minutes before meals + need acid for activation)	Acid stable (Taken before or after meals + Not need acid for activation)

3. H₂-Receptor Antagonists

Cimetidine (prototype) - ranitidine (discontinued due to risk of cancer) - famotidine - nizatidine (oral – IV).

Mechanism & Pharmacological Actions

Reversible competitive antagonists of H₂ receptors on parietal cells → Complete inhibition of histamine mediated but partial inhibition of ACh or gastrin mediated HCl secretion.

Potent inhibition of nocturnal or fasting but less inhibition of daytime (meal-stimulated) acid secretion.

Individual H₂ Antagonists

1. Cimetidine: limited use due to its enzyme inhibiting, antiandrogenic & CNS side effects.

2. Ranitidine, Famotidine, Nizatidine developed to minimize adverse effects of cimetidine (More potent than cimetidine)

Adverse Effects of H₂ Blockers (Mostly with cimetidine)

1. Headache, diarrhea or constipation.
2. Tolerance, rebound hyperacidity & recurrence (upregulation of H₂ receptors).
3. CNS: sedation, confusion, hallucination (mostly with cimetidine especially in elderly, renal or hepatic dysfunction).
4. Antiandrogenic, ↑ prolactin, gynecomastia, galactorrhea, impotence (mostly with cimetidine).
5. Enzyme inhibition → drug interactions (mostly with cimetidine).
6. Hypotension & bradycardia if given rapidly IV in intensive care (Block cardiac H₂ receptors).

Ranitidine (discontinued)	Nizatidine 150 mg	Famotidine 20 mg
Less enzyme inhibiting & antiandrogenic effects.	No enzyme-inhibiting or antiandrogenic effects.	No enzyme-inhibiting or antiandrogenic effects.
Anti H. pylori effect (most efficient).		The most potent Cytoprotective effect.
50% bioavailability (↑hepatic first pass effect)	100% bioavailability (no 1st pass effect)	50% bioavailability. 50% decomposed by gastric acidity.
Eliminated by hepatic metabolism → renal excretion of the drugs or their metabolites → dose reduction if ↓ Creatinine clearance or severe liver disease.		

Advantages of PPIs and P-CABs over H2 antagonists:

1. Higher efficacy → better symptomatic relief & higher response rates.
2. More prolonged effect (= 24hs → given in single daily dose; versus twice daily dosing with H2 antagonists).
3. More effective in anti H. pylori regimen.

Indications of Antisecretory Drugs

- 1- Peptic ulcer: PPIs preferred especially in severe cases. H2 antagonists in mild or moderate cases (healing is delayed for 8 weeks & recurrence is common).
- 2- Anti- H. Pylori regimens: PPIs preferred.
- 3- GERD: PPIs preferred. H2 antagonists are used to cover the nocturnal HCl secretion.
- 4- Zollinger Ellison syndrome: PPI are preferred. High dose of parenteral preparations are given early to cover vomiting period.
- 5- Stress ulcer: to ↓ risk of bleeding.
- 6- Acute gastritis in children & gastric erosions (Parenteral)
- 7- Bleeding oesophageal varices: to prevent recurrence (Parenteral)
- 8- To avoid aspiration pneumonia during surgery e.g. obstetric anesthesia before caesarian section (Parenteral)

Mucosal protectives (cytoprotectives)

	1. Sucralfate	2. Misoprostol	3. Colloidal Bismuth
Mech.	It is an aluminium salt of sulfated sucrose. In acid medium → -ve sulfate groups bind +ve proteins in the ulcer base → protective barrier (also inactivates bile and pepsin) ↑ secretion of bicarbonate, mucus & PGs.	Synthetic PGE1 analog cytoprotective through: ↓ HCl secretion by ↓ histamine-stimulated cAMP production. ↑ mucosal blood flow ↑ secretion of mucus & bicarbonate.	Anti- H pylori activity (main effect). Mucosal protective: Chelates proteins in ulcer base → protective coat against acid & pepsin (inactivates pepsin) ↑ secretion of bicarbonate, mucus & PGs.
Uses:	Not used - Stress ulcer prophylaxis (mainly). Antacids or H2 blockers and some drugs e.g. tetracycline are given 1- 2 hour apart.	Selective for NSAID induced gastric ulcer if PPIs fail. Prophylactically with NSAIDs or corticosteroids.	Anti- H. pylori therapy
ADRs	1. Nausea, vomiting, dry mouth. 2. Flatulence, constipation. 3. Al+3 toxicity: chelates phosphates → osteomalacia & encephalopathy in renal diseases.	1. Diarrhea & colic (most common). 2. Vaginal bleeding & uterine contractions (CI in pregnancy → abortion).	1. Teeth & tongue discoloration - black stools (confused with bleeding). 2. Encephalopathy and neurotoxicity especially in renal disease → not give >2 months & not restarted within 1 year.

Anti H pylori therapy

Eradication of H pylori infection results in rapid healing & ↓ recurrence.

It is achieved by combination therapy with **High efficacy anti- H. pylori Triple therapy.**

Triple therapy: 10-14 days: Lansoprazole + Clarithromycin + Amoxicillin (or metronidazole in patients allergic to amoxicillin).

Quadruple therapy: 10–14 days: PPI + metronidazole 250 mg + bismuth subsalicylate 300 mg + tetracycline 500 mg four times daily.

Sequential therapy: PPI + amoxicillin 1 g twice a day for 5 days followed by PPI + clarithromycin 500 mg and tinidazole/ metronidazole 500 mg twice a day for 5 days

Or: PPI + amoxicillin 1 g twice a day + levofloxacin 250 or 500 mg twice a day for 10 days

Followed by 4-6 weeks: PPI therapy to allow ulcer heal.

N.B.: PPIs may be substituted by P-CABs in Peptic acid diseases and eradication of H. Pylori

Antacids

Antacids are weak bases that reduce gastric acidity by neutralizing HCL

Pharmacological Actions

1. Neutralize gastric acid (duration 1-2 hrs).
2. Decrease peptic activity (pepsin is inactive at pH > 4.5).
3. Reduce H. pylori colonization.
4. ↑ mucosal PGs production (promote defense mechanism).
5. Adsorbent (to pepsin, bile & toxins), astringent (Precipitate surface proteins) and demulcent (soothing) effects (with Al⁺³ & Mg⁺² salts only).

Therapeutic Uses (taken 1 and 3 hours after meals & at bedtime)

1. Rapidly relieve pain & help healing of peptic ulcer.
2. Relieve dyspeptic symptoms.
3. GERD (rapid pain relief).

Drug Interactions: Antacids can affect rates of dissolution, absorption, bioavailability and renal elimination of many drugs through:

1. Changing gastric or urinary pH.
2. Delaying gastric emptying.
3. Chelation: Al⁺³ salts can form insoluble complexes with fluoroquinolones, tetracyclines, digoxin and iron salts → ↓ absorption (do not give antacids within 2 hrs from administration of these drugs).

Types of Antacids

Local Antacids	Systemic Antacids
Al+3 Hydroxide Disadvantage: constipation.	Na HCO₃ Advantages
Mg+2 Salts Disadvantage: diarrhea.	- Rapid onset. - Potent ($\uparrow$ pH > 7).
Ca+2 carbonate Disadvantages 1. Constipation. 2. Rebound hyperacidity (Ca+2 $\rightarrow$ $\uparrow$ acid secretion) $\rightarrow$ $\downarrow$ pain relief & healing. 3. Milk-alkali syndrome (systemic alkalosis - hypercalcemia). 4. $\uparrow$ CO ₂ $\rightarrow$ flatulence & eructation.	Disadvantages 1. Systemic alkalosis on long term use. 2. Rebound hyperacidity. 3. Na+ content is dangerous in HF, hypertension or renal disease. 4. $\uparrow$ CO ₂ $\rightarrow$ flatulence, eructation. 5. Alkaline urine $\rightarrow$ renal stones. N.B.: Sodium bicarbonate is added for effervescent formulations or in immediate release PPIs preparations (e.g. omeprazole).

Choice of antacids

- Al+3 hydroxide & Mg+2 salts do not result in CO₂ release or systemic alkalosis thus they are the most commonly used antacids, alone or combined (e.g. Magalدارate - to neutralize the effects of each other on bowel habit).
- Antacids should be cautiously used in elderly & renal impairment since:
 - Al+3 antacids $\rightarrow$ chelate phosphates $\rightarrow$ osteomalacia, hypophosphatemia, encephalopathy & $\uparrow$ risk of Alzheimer dementia.
 - Mg+2 antacids $\rightarrow$ CNS depression.

Precautions to Avoid NSAIDs-induced Ulcer

1. Use low doses and avoid combinations of NSAIDs ($\uparrow$ toxicity without added benefit).
2. Use topical preparations whenever possible.
3. Co-administer PPIs or misoprostol.
4. Avoid combination with corticosteroids except in selected cases.
5. Use cautiously in the elderly.

MCQs:

1- Which of the following is NOT an advantage of proton pump inhibitors over H2-receptor antagonists

- A. Better symptomatic relief and higher response rates.
- B. More prolonged effect with once daily administration.
- C. More effective in anti H. pylori regimen.
- D. Less risk of gastric carcinoma and infections.

2- Which of the following explains the use of lansoprazole as enteric coated tablets

- A. It is a weak base destroyed by gastric acidity.
- B. It is non ionized in acid medium of stomach.
- C. It prolongs the drug's duration of action.
- D. It is rapidly metabolized by gastric enzymes.

3- Which of the following is correct concerning the Pharmacokinetics of proton pump inhibitors

- A. Acid production in parietal cell canaliculi prevents their activation.
- B. Repeated administration decreases their bioavailability.
- C. Food increases their bioavailability thus taken after breakfast meal.
- D. Taken 30 minutes before meal to be activated by food induced acid.

4- Which of the following is correct concerning Pharmacokinetics of H2-receptor antagonists

- A. Most of them has low to moderate bioavailability except nizatidine.
- B. 50 % of the dose of famotidine is lost by hepatic first pass effect.
- C. 50 % of the dose of ranitidine is destroyed by gastric acidity.
- D. They are contraindicated in mild renal dysfunction due to renal excretion.

5- Which of the following differentiates Vonoprazan from Lansoprazole

- A. Vonoprazan has longer duration of action.
- B. Vonoprazan is directly active on proton pump.
- C. Vonoprazan needs gastric acid for activation.
- D. Vonoprazan has lower efficacy on proton pump.

6- Which of the following drugs blocks basal as well as meal stimulated gastric acid secretion without affecting cholinergic, histaminergic or gastrin receptors

- A. Famotidine.
- B. Misoprostol.
- C. Omeprazole.
- D. Pirenzepine.

7- A 45-year-old man presents with a history of recurrent burning pain in the epigastric region, especially after meals. He also reports occasional bloating and indigestion. After an upper GI endoscopy, he is diagnosed with peptic ulcer disease. His physician prescribes pantoprazole to manage his condition. Which of the following is the most likely mechanism of action of the drug prescribed

- A. Inhibits H₂ receptors on parietal cells.
- B. Increases prostaglandin production in the stomach.
- C. Inhibits the enzyme urease in *Helicobacter pylori*.
- D. Inhibits parietal H⁺/K⁺ ATPase.

8- 50-year-old woman presents with complaints of chronic heartburn and regurgitation, especially at night. She is diagnosed with gastroesophageal reflux disease (GERD). The physician prescribes famotidine for her condition. Which of the following is the primary mechanism of action of famotidine

- A. Blocking muscarinic receptors on parietal cells.
- B. Inhibition of the proton pump.
- C. Blocking histamine H₂ receptors in the stomach.
- D. Neutralization of gastric acid.

9- A 32-year-old male with a history of *Helicobacter pylori* infection and peptic ulcer disease is started on triple therapy, which includes a PPI, amoxicillin, and clarithromycin. After 14 days of treatment, he returns for follow-up. PPI is continued. What is the primary goal of using a proton pump inhibitor in this regimen

- A. To promote mucosal healing by reducing acid secretion.
- B. To neutralize gastric acid.
- C. To enhance the antibacterial effect of clarithromycin.
- D. To directly kill *Helicobacter pylori*.

10- A 45-year-old woman with peptic ulcer disease is treated with misoprostol. She is counseled regarding potential side effects of the medication. Which of the following is a common adverse effect of misoprostol

- A. Headache
- B. Diarrhea
- C. Constipation
- D. Hypertension

11- A 50-year-old man with a history of stress ulcer is prescribed sucralfate as part of his treatment regimen. The physician explains the mechanism of action to the patient. Which of the following best describes the action of sucralfate in this case

- A. Inhibits H₂ receptors on parietal cells.
- B. Stimulates acid secretion.
- C. Acts to neutralize gastric acid.
- D. Forms a protective barrier over the ulcer.

12- Which of the following is NOT correct concerning misoprostol

- A. PGE analog that increases mucosa blood flow, mucus & bicarbonate.
- B. Contraindicated during pregnancy as it may induce abortion.
- C. Diarrhea and abdominal cramps its most common adverse effects.
- D. A cytoprotective agent used in all types of peptic ulcer if H₂ blockers fail.

13- Which of the following is NOT true concerning bismuth subsalicylate

- A. It acts as mucosal protective agent.
- B. It is used in quadruple anti- pylori regimen.
- C. It should be given for more than 4 months.
- D. It may cause neurotoxicity & encephalopathy.

14- Which of the following is the major advantage of potassium competitive acid blockers

- A. They have favorable pharmacokinetic profile than proton pump inhibitors.
- B. They have shorter duration of action than H₂ receptor blockers.
- C. They are prodrugs that gives them longer duration of action.
- D. They are used as much smaller doses than proton pump inhibitors.

15- Which of the following phases of acid secretion is most likely affected by H₂ receptor antagonists

- A. Meal stimulated acid secretion.
- B. Nocturnal acid secretion.
- C. Both nocturnal and meal stimulated phases.
- D. Gastrin stimulated acid secretion.

16- Which of the following medications is mostly included in anti H pylori regimens

- A. Clarithromycin.
- B. Bismuth subcitrate.
- C. Doxycycline.
- D. Misoprostol.

17- Which of the following mucosal protective agents has anti H pylori effect

- A. Sucralfate.
- B. Misoprostol.
- C. Bicarbonate.
- D. Bismuth subcitrate.

18- Which of the following is the major side effect of Magnesium hydroxide

- A. Systemic alkalosis.
- B. Diarrhea.
- C. Constipation.
- D. Rebound hyperacidity.

19- 50-year-old man who has been diagnosed with Helicobacter pylori-induced peptic ulcer disease is started on triple therapy that includes a proton pump inhibitor, clarithromycin, and metronidazole and an antiemetic domperidone. He complains of headache, dizziness, somnolence and nausea during the course of the treatment. What is the most likely cause of the patient's symptoms

- A. Clarithromycin.
- B. Metronidazole.
- C. Proton pump inhibitor.
- D. Domperidone.

20- Systemic alkalosis could be an adverse effect of long term use of which of the following antacids

- A. Magnesium oxide.
- B. Magnesium hydroxide.
- C. Aluminum hydroxide.
- D. Sodium bicarbonate.

21- Which of the following antacids is mostly preferred for hyperacidity

- A. Magnesium hydroxide.
- B. Aluminum hydroxide.
- C. Both Mg and Al salts together.
- D. Sodium bicarbonate.

22- Which of the following antacids has the most rapid and potent effect in hyperacidity

- A. Sodium bicarbonate.
- B. Aluminum hydroxide.
- C. Magnesium hydroxide.
- D. Magnesium oxide.

23- A 25-year-old woman with a known history of gastritis and recurrent epigastric pain is prescribed pantoprazole, a proton pump inhibitor. When should pantoprazole be taken for optimal effect

- A. Immediately after meals.
- B. At bedtime.
- C. 30 minutes before meals.
- D. With food.

24- A 40-year-old man is diagnosed with gastroesophageal reflux disease (GERD) and is prescribed omeprazole, for treatment. He asks about the potential long-term risks of this medication. Which of the following is a potential long-term complication of chronic PPI use

- A. Gastrointestinal bleeding.
- B. Osteoporosis.
- C. Peptic ulcers.
- D. Bradycardia.

25- A 45-year-old male with peptic ulcer disease is prescribed a potassium-competitive acid blocker (P-CAB). He asks how this medication works to reduce stomach acid production. Which of the following best describes the action of P-CABs

- A. They directly inhibit the secretion of gastric acid by parietal cells
- B. They block the binding of histamine to H₂ receptors
- C. They block the potassium-binding site of the proton pump
- D. They increase the production of bicarbonate in the stomach

26- Which of the following is the primary mechanism of action of Vonoprazan

- A. Inhibition of histamine H₂ receptors.
- B. Neutralization of gastric acid.
- C. Stimulation of gastric prostaglandin production.
- D. Inhibition of H⁺/K⁺ ATPase reversibly.

27- Which of the following drugs is an example of a potassium-competitive acid blocker (P-CAB)

- A. Omeprazole.
- B. Pantoprazole.
- C. Vonoprazan.
- D. Misoprostol.

28- Which of the following is a key advantage of potassium-competitive acid blockers (P-CABs) compared to proton pump inhibitors (PPIs)

- A. P-CABs are more effective antisecretory drugs regardless food intake.
- B. P-CABs have a longer duration of action.
- C. P-CABs are less effective in treating *Helicobacter pylori* infections.
- D. P-CABs require less frequent dosing than PPIs.

29- A 60-year-old woman with gastroesophageal reflux disease (GERD) is started on Vonoprazan. After several weeks, she complains of mild diarrhea and abdominal discomfort. She said that she used an H2 blocker without being prescribed. What is the most likely cause of her symptoms

- A. The potassium-competitive acid blocker (P-CAB).
- B. Excessive acid suppression leading to bacterial overgrowth.
- C. Hypersensitivity reaction to the medication.
- D. The use of an H2 receptor antagonist in conjunction with P-CAB.

30- Which of the following drugs mediates its ulcer protective effect by increasing mucosal blood flow through a vasodilator effect

- A. Sucralfate.
- B. Misoprostol.
- C. Bismuth subsalicylate.
- D. Aluminum hydroxide.

Answers:

- | | |
|-------|-------|
| 1) D | 16) A |
| 2) A | 17) D |
| 3) D | 18) B |
| 4) A | 19) C |
| 5) B | 20) D |
| 6) C | 21) C |
| 7) D | 22) A |
| 8) C | 23) C |
| 9) A | 24) B |
| 10) B | 25) C |
| 11) D | 26) D |
| 12) D | 27) C |
| 13) C | 28) A |
| 14) A | 29) B |
| 15) B | 30) B |

Pathology of GIT tumors

Tumors of esophagus

Benign tumors of the esophagus are less common than malignant tumors.

The most common are **leiomyomas**.

Malignant tumors may be either

Primary Tumors

1- Epithelial → Carcinoma (most common).

2- Mesenchymal → (Sarcoma) e.g. Leiomyosarcoma (rare).

Secondary Tumors

(Rare in the esophagus)

Clinical features of esophageal carcinoma

They present by esophageal obstruction, dysphagia, bleeding, or rarely, paraneoplastic syndrome.

Diagnosis is usually made by imaging techniques and endoscopic biopsy.

Prognosis of esophageal carcinoma is usually poor (5 year cure rate about 10%).

Carcinoma of esophagus

Squamous cell carcinoma - More prevalent worldwide 50% in middle 1/3

Risk factors: long-standing esophagitis, achalasia, smoking, alcohol, diet (low vitamins and zinc), genetics

Adenocarcinoma - More common in USA - More in lower 1/3., Occurs on top of Barrett esophagus, preceded by dysplastic changes.

Gross Pathology

- 1. Exophytic (Polypoid or Fungating).** Growth filling the lumen (Most common type).
- 2. Ulcerative type, which invade deeply into the surrounding structure.** With raised reverted edge, rough necrotic floor and indurated base.
- 3. Endophytic - Infiltrating (Schirrhous) tumor** encircling the tube, also called (Annular) & cause constriction (Is the least frequent type).

Histopathology

1. Squamous Cell Carcinoma: Occur in >90 % of cases.

The tumor is composed of islands of neoplastic squamous cells. with epithelial cell nests in differentiated tumor.

2. Adenocarcinoma (5%): Originating in the cardiac end of the esophagus from dysplastic mucosa in the setting of Columnar Metaplasia (Barrett's oesophagus).

3. Undifferentiated Carcinoma: lacking squamous features & accompanied by lymphocytic infiltration.

Effects & Complication

- Dysphagia.
- Perforation of ulcer with fatal haemorrhage.
- Hematemesis.
- Spread:
 - Locally.
 - Lymphatic.
 - Blood, in late stage.

Gastric tumors

Benign Neoplasms

Epithelial polyps are rare in the stomach. Three types occur:

- 1- Hyperplastic polyps (70%),
- 2- Fundic gland polyps (20%),
- 3- Adenomatous polyps (10%). Shows dysplasia.

The risk of carcinoma is moderate in adenomatous polyp, slight in hyperplastic polyp, and nil in fundic gland polyps.

Gastric polyps appear as small pedunculated lesions often multiple, that can be removed endoscopically, Large polyps are very rare.

Mesenchymal Neoplasms : Benign mesenchymal neoplasms are uncommon.

Malignant Neoplasms

Gastric Adenocarcinoma

Sex: male > female - **Geography:** Japan, Chile, Iceland - **Age** > 50.

Predisposing factors

- Related to the socio-economic factors.
- Genetic factors.
- Dietary factors: foods that may Generate nitrites (smoked foods, pickled vegetables and excessive salt intake).
- Blood group A = 20% increased risk , Adenomatous polyp , Pernicious anemia (atrophic gastritis type A) , Chronic gastritis associated with Helicobacter pylori infection.

Precancerous lesions in the stomach (Dysplastic Lesions)

- 1- Intestinal metaplasia due to: Chronic Gastritis, Infection with H. pylori
- 2- Peptic ulcer of the stomach.
- 3- Adenomatous polyp.
- 4- Chronic Gastritis:
 - H. pylori associated Gastritis .
 - Chronic Autoimmune Atrophic Gastritis.
 - Intestinal metaplasia

Pathological changes Related To H. Pylori

H. Pylori infection → Gastritis → Atrophic gastritis → Intestinal metaplasia → Dysplasia → Cancer.

Site:

The favoured location is the **Pyloric Region** (50%).

Followed by lesser curvature (25%).

but can occur in any part of the stomach.

Grossly it may appear as:

- **Early gastric cancer:** defined as gastric carcinoma restricted to the mucosa and submucosa - Appears as a small, flat mucosal thickening that may have a minimal polypoid and ulcerative component.
- Late gastric cancer... defined as a gastric carcinoma that has invaded the muscle wall, It may present in various ways:
 - (1) Exophytic (Polypoid & Fungating): It form friable projecting intraluminal mass.
 - (2) Ulcerative: the ulcer is more than 5 cm, with everted edge, necrotic floor and firm base.
 - (3) Diffusely infiltrating lesion (Flat or depressed): that causes thickening and contraction of the stomach wall with relatively little mucosal involvement without forming localized tumor mass. (linitis plastica, or leather bottle stomach).

Histological Classification

Lauren Classification:

Intestinal Type → (53%).

Diffuse Type → (33%).

Unclassified → (14%).

1- Intestinal-type:

Risk factors: diet (nitrites, smoked food, increased salt), chronic gastritis (H. pylori), altered anatomy after resection.

On top of intestinal metaplasia.

Histopathology: Glandular morphology.

2- Diffuse-type:

Undefined risk factors (no known relation to H. pylori)

Signet cell morphology

Spread of gastric carcinoma

A) Direct spread through the wall (Staging 1-4)

Then to the neighboring structures (esophagus, duodenum, omentum, liver & rarely to the spleen).

B) Lymphatic spread to the lymph nodes along the Greater & Lesser Curvatures, Para Aortic & Supra-clavicular Lymph Nodes.

C) Blood stream to the Liver, Lung, Bone, Brain.

D) Transcoelomic implantation over the peritoneal surface to the Ovary (Krukenberg tumor).

Tumors of the Colon and Rectum

Tumors of all types are not very common in small intestine.

Epithelial tumors are Common in large intestine.

Tumors of large intestine

Benign Tumors

1. Familial Polyposis Syndrome.

2. Adenoma

- Tubular.
- Villous.
- Tubulo-villous.

Malignant Tumors

1- Carcinoma (Adenocarcinoma).

2- Lymphoma & sarcoma.

Neoplastic adenomatous polyp of the colon

Benign Tumor arising from surface epithelium, in the form Of Polypoid Non-capsulated Mass.

Age > 60.

More common in large intestine.

There is a well-defined familial predisposition.

Predispose to colorectal Carcinoma

M/P:

1- Tubular adenomas (pedunculated):

- The most common account for over 90% of colonic adenomas.
- Small, Multiple with 10–20 lesions present in some patients.
- Pedunculated with a well-defined stalk .
- Histologically, a tubular adenoma is composed of benign neoplastic glands bunched together above the muscularis.
- The epithelial cells are dysplastic (hyperchromatic and stratified and show loss of normal mucin content)
- Closely packed epithelial tubules, lined by dysplastic cells.

2- Villous adenomas – (sessile):

- Uncommon, comprising less than 10% of colonic adenomas.
- They commonly occur in older individuals as a single large, sessile lesion have a broad base of attachment to the mucosal surface without a defined stalk.
- Histologically, villous adenoma is composed of neoplastic proliferation of colonic epithelial cells organized into long finger-like papillary or villous processes.
- Large > 2cm, Solitary, Sessile.
- Dysplasia & Cancer occurs in high percent (40%).

3- Tubulo-villous adenomas:

- Pedunculated, Composed of a mix of tubular & villous areas.

Villous Adenomas	Tubular Adenomas
Less common	More common
Large > 2cm	Small < 1cm
Solitary	Multiple
Sessile	Pedunculated
Dysplasia & cancer is high (40%) in adenomas > 4cm in diameter.	Dysplasia & cancer is rare.

G\P:

1- Pedunculated With neck

2- Sessile without neck

Colonic adenomas are associated with genetic mutations identical to those seen in colon cancer.

The number of mutations in adenomas is less than in carcinoma but increases as the adenoma enlarges and becomes more dysplastic.

Inherited Polyposis Syndromes

(Adenomatous Polyposis Coli APC - Familial Adenomatous Polyposis (FAP), - Gardner Syndrome. - Turcot Syndrome. - Hereditary Non-polyposis Colorectal Cancer Syndrome (HNPCC)

Familial Adenomatous Polyposis (FAP)

- Uncommon hereditary Autosomal dominant disorders
- Average onset is 25 years old
- Adenomatous polyps, mostly colorectal.
- Characterized by presence of hundreds of colonic polyps.(A minimum number of 100 is required for diagnosis.)
- Most polyps are tubular adenoma.
- The genetic defect is on chromosome 5q21, APC (Adenomatous Polyposis Coli) tumor suppressor gene
- Precancerous: virtually 100% will develop cancer within 10 to 15 years unless a prophylactic colectomy is performed.

Malignant colonic tumors

I- Epithelial Adenocarcinoma and mucoid adenocarcinoma

II- Mesenchymal - Gastro-intestinal Stroma Tumor (GIST) - Lymphoma Colonic.

A great majority (98%) of all cancers in the large intestine are adenocarcinoma.

Prevalent in males twice as common as in females.

Age of 50-70 years.

Almost always arises from adenomatous polyp

Adenocarcinoma

Predisposing Factors:

Adenoma of the colon.

Familial polyposis.

IBD (Ulcerative colitis, Chron's disease).

Irradiation.

Mutations in specific Genes (e.g. DNA mismatch repair genes, APC, k-ras, p53).

Risk factors

low fiber, high fat Diet

Decreased Vit A, C, E

Increased meat consumption

- **Survival:** depends on stage (depth of invasion and node metastasis)
- **Incidence:** Primarily occurs in patients between 60 and 79 years of age
- **Ratio of male to female** patients 2:1 for rectal cancers, 1:1 before 50).

Gross appearance:

Sites:

Rectum and distal sigmoid (50%)

Caecum & Ascending colon (25%)

Descending colon & proximal sigmoid (20%)

The Remainder 5% are Scattered Elsewhere In The Colon.

Shape:

1. Exophytic (Fungating & Polypoid) usually in the proximal colon and caecum and ascending colon, Obstruction is uncommon, due to large caliber of caecum & ascending colon.
2. Ulcerating: With raised everted edge, necrotic floor, hard base.
3. Infiltrating (Annular).

Histopathology

Which may be:

1. Well-differentiated adenocarcinoma: Invasive Malignant acini of variable degree of differentiation, with muscle invasion.

2. Muroid (mucinous) carcinoma (Subtype of adenocarcinoma):

A- Mucinous adenocarcinoma: characterized by abundant extracellular mucin.

B- Signet ring cell carcinoma: malignant cells show intracytoplasmic mucin pushes the nuclei eccentrically .

3. Undifferentiated Schirrhous carcinoma.

- Glands have lost polarity, disorganized and adhere to each other
- Epithelial cells are hyperchromatic, tightly packed, and have a higher nucleus-to cytoplasm content (appear more malignant).
- The malignant acini are separated by desmoplastic stroma.
- Tumor invades deeply into colonic mucosa
- Tumor cells may continue to produce mucin; in 10-15% of patients, mucin lakes can be seen.

Course: tumor invades bowel wall and lymphatics/blood vessels with metastasis to lymph nodes, liver, lungs, and bones.

25% of patients have metastatic disease at presentation.

Diagnosis is based on:

o Endoscopy and biopsy.

o Increased serum CEA which is more valuable in follow up after surgical removal of a tumor.

Intestinal Carcinoid

Carcinoid tumors arise from neuroendocrine cells present in the mucosa throughout the gastrointestinal tract.

Grossly,

- Carcinoid tumors are firm and yellow.
- They arise in the submucosa, elevating and ulcerating the mucosa and locally infiltrating the wall.
- Multiple lesions are present in 25% of cases.

Microscopically

- Sheets of monotonous, polyhedral cells Separated by Fibrovascular stroma with deeply stained nuclei.
- Stained with silver or by IHC for chromogranin or neurone specific enolase.

Gastrointestinal stromal tumor (GIST)

Incidence: Gastrointestinal stromal tumors are the most common mesenchymal neoplasms of the gastrointestinal tract.

Origin: GISTs arise in the smooth muscle pacemaker interstitial cell of Cajal.

Gross:

large, bulky, intramural mass with tannish brown cut section with hemorrhage, necrosis and cystic softening.

Microscope:

Mesenchymal tumor usually centered in the muscularis propria

There are two types:

1-Spindle cell GISTs: intersecting fascicles of plump spindled cells with eosinophilic cytoplasm within variably hyalinized or edematous stroma.

2- Epithelioid GISTs: rounded epithelioid cells.

Foodborn Infection

Foodborn illness: are infections caused by a variety of microorganisms mainly Salmonella, Clostridium perfringens, Campylobacter, and Staphylococcus aureus norovirus.

Salmonella is a member of Enterobacteriaceae family, it is the causative agent of:

Enteric Fever (typhoid) or **Salmonellosis**, there is bacterial **invasion of bloodstream** by Salmonella.

Acute gastroenteritis, resulting from a **food-borne** infection mainly contaminated **eggs** and intoxication.

Poultry are the primary reservoirs

Salmonella

Morphology: Gram negative bacilli, **motile**, non spore forming.

Culture:

- **Facultative** anaerobes, grow well at 37°C after 24 hours.
- **MacConkey agar: non-lactose fermenters (pale yellow colonies).**
- Other **selective media** used are **SS agar** and **deoxycholate citrate agar (DCA)** and broth selenite and tetrathionate are used for its isolation.

Biochemical reactions:

Partial sugar fermentation with the production of **acid and gas** by most species **except S.typhi** which produce **acid only**. All are **urease and oxidase negative** and **H₂S positive**.

Antigenic structure:

- 1- **O (somatic) antigens** which subdivide the Salmonellae into **groups**.
- 2- **H antigen (flagella) antigens** which determine the **type** within the groups.
- 3- **Virulent antigen (Vi Ag)** is **anti-phagocytic** and is present **only in S. typhi**.

Pathogenesis

It is due to endotoxins and exotoxins.

The exotoxins such as **enterotoxins** is responsible for diarrhea and body fluid loss .

Antigens

Resistance:

Killed by **heating at 60°C for 30 min and antibiotics** and other chemotherapy.

Resistant to the action of **bile salts**.

Diseases caused by Salmonella:

A) Enteric fever

Enteric fever is caused by **S. typhi** and the **milder** paratyphoid fever is caused by **Paratyphi** A, B and C.

Enteric fever is defined as **generalized infection of reticuloendothelial system** and lymphoid tissue accompanied by **sustained fever** and bacteremia.

Enteric fevers are **endemic** in Egypt.

Pathogenesis:

Enteric fever is a **strictly human** disease, from **case or carrier stool and urine**.

The organisms enter the human digestive tract with contaminated food or drink e.g **raw shell fish ,milk products and raw vegetables**.

The organism adhere to the **mucosa of small intestine** by **pili** then penetrate to submucosal layer causing no lesion.

In lymphoid tissue the organism **invade macrophages** where bacterial multiplication occurs.

They are transported to **mesenteric lymph nodes** and lymphatics, released into the **bloodstream** result in **bacteremia** and are carried to many organs such as **bone marrow, kidneys**, intestine, **spleen and liver** where replication of salmonella inside the macrophages occurs.

After multiplication in **reticuloendothelial system** it is released into bloodstream and results **signs of septicemia mainly fever**.

The organism is transported to **the gall bladder** to be excreted with bile.

Salmonella multiply in intestinal lymphoid tissue and are excreted in **stools at the second week**.

The incubation period of the disease is **10-14 days**.

The fever is **gradually elevated** during the first week then become steady during the second week.

There is **enlargement of spleen and liver**.

The white cell count is **normal or low**.

Salmonella **excretion** by patients may **continue long after clinical cure**.

Outcome:

5% of patients cured from typhoid **remain carriers** for months or even years.

Urinary carriers are common in Egypt because of bilharziasis.

Laboratory diagnosis: Specimens are collected according to the **stages** of the disease.

A- Isolation of Salmonella

- 1- **During the first week by doing blood culture** in fluid media and then subculture on solid media mainly MacConkeys media at 37°C.
- 2- **Second week** , Salmonella is isolated **from stool** and after the second week from urine by direct culture on selective media.
- 3- **After the second week** Salmonella is isolated **from urine** by direct culture on solid media .

B- Serological diagnosis by Widal agglutination test:

This test demonstrates agglutination of **serum** antibodies of the patients against **O-somatic antigens** of Salmonella and **H-flagellar** antigens of S.typhi and Paratyphi A,B,C.

Antibodies **rise sharply** during the **second and third weeks** of enteric fevers.

Demonstration of **a rising titers** in antibodies in **the paired** samples indicates **acute recent infections**.

The results should include the titer of both O and H antibodies and Vi antibodies.

The presence of **IgM** is indicative of **acute recent infections**.

Interpretation of Widal test:

- 1- infection is present.
 - 2- infection with this type of Salmonella.
 - 3- **High titer** of antibody to the **Vi antigen** occur in some carriers.
 - 4- **High titer** of antibodies to **O antigens ($\geq 1:160$)** suggests that **active infection** is present.
 - 5- More than one **H antigen ($\geq 1:160$)** suggest **recent** infection with this type of Salmonella and recent immunization.
- With **old** immunization or past infection **O antibodies disappear** and only antibodies to H antigens remain.

Widal test has many **limitations** as a diagnostic serological test for enteric fever.

Several factors need to be considered for interpretation:

- **False positive** results may occur: in **endemicity** of enteric fever in an area of investigation, **immunization** with any typhoid vaccine or a **previous infection** or infection with **other related members** of Enterobacteriaceae.
- **False negative** in case of administration of **antibiotics** and if done in **early stage of disease**.
- **Antibiotics** may be **not effective** on **Salmonella carriage** due to **hidden carriage site**.

Prevention and Control by

1- Food handlers are examined to **diagnose carriers**, treat them or excluding them from handling food.

2- Immunization: several typhoid vaccines are available for use :

a. Parenteral heat-phenol-inactivated vaccine contains S.typhi and paratyphi A,B,,C hence the name **TAB vaccine**.

The vaccine is given as **2 subcutaneous** (SC) doses.

This is the current vaccine used in Egypt.

b. Other vaccines include: **acetone-inactivated parenteral vaccine**,oral live attenuated vaccine for S. typhi. Living vaccine should not be given to children younger than 6 years of age. Vi capsular polysaccharide (ViCPS) is a newly licensed capsular polysaccharide vaccine for parenteral (intramuscular) use.

No typhoid vaccine is 100% effective and is not a substitute for good hygiene

Salmonella food poisoning

Is caused by

A- Salmonella typhimurium foodborne toxic infections. Symptoms (diarrhea, vomiting, fever) appear **after 12-24 hours** following ingestion of **meat contaminated with rat excreta** which contain **large number** of organisms. It last for 2-5 days, spontaneous cure usually occurs.

B- Salmonella enteritidis infection or **Egg-associated** salmonellosis

The egg can be inside **perfectly normal**.

Incubation period from **12 to 72 hours** after consuming a contaminated food.

The illness usually **lasts 4 to 7 days**, and most persons recover without antibiotic treatment.

Campylobacter

12-15 species are associated with human disease:

- Campylobacter **jejuni** (**poultry**; humans)
- Campylobacter coli (porcine; humans)
- Campylobacter **fetus** (**cattle**; sheep).

Morphology: They are **gram negative curved bacilli** (S-shaped or "**gull-winged**" **appearance**), non-spore forming, have a characteristic rapid **darting motility** by means of long sheathed **polar flagellum** at one (polar) or both (bipolar) ends.

Motility **slows quickly** in wet mount microscopy preparation due to aerotoxicity (**oxygen sensitivity**).

Culture:

A- **Micro-aerophilic** and **capnophilic** (CO₂ :10%)

B- **Thermophilic** Incubation temperature is at **42°C** as most strains are (natural body temperature of **avian species**).

C- **Selective media** with added antibiotics as **Skirrows medium**.

Biochemical reactions: **Oxidase positive**, catalase positive, **do not utilize carbohydrates**.

Pathogenesis: Virulence factors include:

- a- Cellular components: **Endotoxin(LPS)**, **flagellum**, adhesins and invasins.
- b- Extracellular components: **Enterotoxins (LT)** and **cytotoxins**.

Transmission: **feco-oral route**.

Campylobacter is usually acquired from **food** (specially **poultry**), milk, or **contact with infected animals** (e.g. dogs).

Children **under 5 years** and **young adults** are the most frequently affected age groups.

Incubation period of 1-7 days with **acute onset**, abdominal cramps, nausea but not vomiting.

The bacteria damage the **mucosal surfaces of jejunum**; ileum and **colon**.

Stools may contain **mucus or gross blood**.

Generally **self-limiting**, symptoms may last for a week or longer. Bacteremia is common and Campylobacter is **isolated from blood**.

Important late onset complications following gastroenteritis include:

- **Guillian-Barre Syndrome.**
- Reiter's syndrome.
- reactive arthritis.
- **spontaneous abortion.**

Laboratory diagnosis:

Specimen collection and processing **stool** should be examined within few hours.

Rectal swab: Care to avoid **oxygen exposure** while taking sample or better a special transport medium is used.

A- Direct examination:

- Gram stained smears of stools can show **curved or S-shaped** bacilli.
- Darting motility in **wet mount** preparation **under dark field** illumination.

B- Culture: organisms are isolated on selective media .Identification of isolated colonies is done on the basis of, biotyping ,**serotyping**, phage typing.

Antibiotic resistance:

Fluoroquinolone resistance has developed rapidly related to unrestricted use and the use of **enrofloxacin in poultry**.

The organism is **inherently resistant to trimethoprim** and most **cephalosporins**.

Brucella

Brucella is the cause of **brucellosis**, a **zoonotic** disease.

It is a virulent, highly infectious bacteria.

They are facultative **intracellular** pathogen surviving in **macrophages**.

There are different species of Brucella which include: Brucella abortus (cattle), Brucella melintensis (goats, sheep), Brucella suis (swine,pigs), Brucella canis (dogs).

Species are differentiated by **production of urease and H₂S, dye sensitivity**, cell wall antigens and phage sensitivity.

Morphology: Gram-negative small bacilli, non-spore forming, **non-motile** and non capsulated.

Culture: Aerobic, **10% CO₂** require enriched media containing **serum** e.g. serum dextrose agar, Colonies appear after **48 to 72 hours**.

Biochemical reactions: **Positive** for catalase, **oxidase ,urease** and reduce nitrates.

Brucellosis (Malta fever or undulant fever): This a true **zoonotic** disease as human to human transmission has not been identified.

The disease is characterized by **acute undulating fever** i.e **bouts of fever** 3-4 weeks then 3-4 weeks without fever.

Transmission:

Direct contact through **skin abrasions ,mucous membrane** with infected animals or animal products (farmers, butchers and veterinarians).

Consumption of **unpasteurized** milk products and meat.

Inhalation of aerosols from infected animal or **laboratory-associated**.

Laboratory diagnosis:

Specimen: blood is taken during acute disease, Other sample according to site of infection such as joint fluid, CSF, pus.

Gram stain is of no significance.

A- Culture: **Blood culture** is the method of choice early in the disease.

Cultures need to be incubated for **up to 4-6 weeks** as they are slow -growing.

It is important to differentiate **Brucella** from **Salmonella** by **urease test** as brucella produce **urease (+ve)** while Salmonella not (-ve).

B- Serology: Is the **main method** of laboratory diagnosis.

Two serum samples is collected 14-21 days apart.

A **wide range of serum dilutions** must be done in order to avoid false negative result due to **prozone effect** that is commonly present in cases of brucellosis **as large amount of antibodies are produced**. Dilutions are done from **1:20 to 1:520**.

ELISA is used to differentiate between specific **IgM and IgG antibodies**.

C- Brucellin test: **intradermal test** based on delayed type hyper-sensitivity.

Interpretation of the result is **difficult in persons frequently exposed** to infection.

Prevention:

Wearing **protective** impermeable clothing, rubber boots, gloves and face masks when exposed to animals.

Pasteurization of milk and treatment of domestic animals.

The use of **live attenuated** vaccines for animals (cattle).

Treatment: using combinations of antibiotics for **4 to 6 weeks**.

Food born poisoning

Staphylococcus aureus: are **catalase positive** Gram-positive cocci.

Staphylococcal enterotoxins **do not change the taste, color or odor of food**.

They are **heat stable for 30 minutes** of boiling and are not inactivated by brief cooking.

The source of food contamination is a **food handlers** carrying the organism in **nose, hands**, or having superficial lesions as **boil (abscess)**.

The common types of food are **carbohydrate rich food** e.g.cake, koshari and **protein rich food** e.g milk and its product.

The incubation period is short ,**1-6 hours**, due consumption of **preformed** enterotoxins.

The food poisoning is characterized by **vomiting** and **watery non bloody diarrhea**.

Laboratory diagnosis: Specimen include **food, vomitus**.

Direct examination: Gram positive cocci in grape like clusters.

Culture is done on **blood agar** to isolate beta (complete) hemolytic colonies and by using selective **mannitol-salt agar**.

Typing of S .aureus is done for epidemiological purposes, in food poisoning outbreaks outbreaks by **phage typing** and **PCR**.

Salmonella food poisoning: under Salmonella Spore forming gram positive bacilli : The gram positive spore forming bacilli include **Bacillus and Clostridium species**.

Bacillus species are **aerobes** while clostridia are **anaerobes**.

1. B.cereus cause two forms of food poisoning

a. The "short-incubation" or **emetic form** of the disease is mediated by **preformed heat stable enterotoxin** usually in **improperly stored, cooked rice**.

It has an incubation period of **1-6 hours**.

It resembles S. aureus food poisoning

b. The "long-incubation" or **diarrheal form**: caused by **heat labile** enterotoxin present in **meat** or vegetables with an incubation period of 8-16 hours.

This type of exotoxin **activate adenylate cyclase**, increase cAMP cause profuse fluid secretion.

In either type, the illness usually lasts for **less than 24 hours** after onset.

2. Clostridium botulinum: is the causative agent of Botulism which is a **rare but fatal** form of food poisoning, is caused by a **potent botulinum toxin**.

Transmission and pathogenesis: **Toxin** is produced within the **contaminated canned food** and **ingested preformed** then becomes **activated by proteases** in the gastric fluid and by **gastric acidity** then absorbed in the intestine and is transported systemically **via the blood stream**.

It binds to **receptors on peripheral nerves**, inhibits the release of **acetylcholine** and lead to **flaccid paralysis**.

Botulinum toxin is rapidly acting and signs and symptoms appear within **12 hours** after ingestion.

Antitoxin is only effective before the toxin binds the neuromuscular junction **within 12 h** after ingestion.

Prognosis is **poor** for patients diagnosed after this time period.

Laboratory diagnosis: Diagnosis depend on clinical manifestations.

The presence of exotoxins in **canned food**, is detected by by ELISA .

Trivalent anti-toxin serum (types **A,B and E**) is given **IV** along with respiratory support are used as treatment.

Antibiotics are **not used** to treat botulism.

Prevention: **Swollen cans** must be discarded.

Other GIT pathogens include

Leptospirosis (Weil's disease, 7-day fever)

Leptospira are spirochetes that live in **stagnant water, lakes**.

Morphology: Spiral-shaped, **motile**, one or both ends are **hooked**, because they are **so thin**, live Leptospira are best observed by **dark-field microscopy**.

Culture: Aerobic, at 30 °C, **need enriched media (Stuart 's media)** to enhance growth that appear after **1-2 weeks**.

It is a **bacterial zoonotic** disease caused by Leptospira.

The infection is commonly transmitted to humans by allowing **fresh water** that has been contaminated by **animal urine** to come in contact with breaks in the **skin, eyes** or with the mucous membranes.

Occupational risk factors include **veterinarians, slaughter house workers, farmers, mine, and sewer workers**.

The disease is **not known** to be spread from **person to person**.

Pathogenesis: Incubation period: **4-14 days**.

It is a **biphasic disease**, the patient is asymptomatic in between phases: **Variable febrile onset**: last for 4-7 days.

There is hepatitis and **jaundice** and the second phase is manifested by **meningitis** (in 40%) with intense **headache and stiff neck, hepatitis, nephritis may recur**.

Laboratory diagnosis: Specimen: Organisms are present in **blood** for **the first 7 to 10 days**, after this period the microorganism can be found in **fresh urine**, kidney biopsy.

Direct examination: Dark-field microscope and PCR.

Culture: Colonies are identified by agglutination test, DNA probes.

Serologic diagnosis: By **ELISA for IgM** and total antibody.

There are no human vaccines, animal vaccines are effective for a few months.

Yellow Fever (YF)

It is a **haemorrhagic fever** caused by yellow fever virus which is a **single-stranded enveloped RNA arboviruses** (arthropod borne virus).

The disease is present in **equatorial countries** in Africa and South America (Amazon basin).

Man usually forms an **accidental, non-normal** host for the virus.

Pathogenesis and disease:

Incubation period: 3-5 days.

Human activity can bring man into contact with **mosquitos** e.g. clearing of forests, agriculture.

The virus multiply in the **reticuloendothelial cells**, then viremia occur to be localized in **liver, spleen, kidneys and bone marrow**.

This is a severe systemic disease in which **hemorrhages**, degeneration of the liver, kidney and heart occur.

Jaundice is a prominent feature due to liver necrosis.

The case-fatality rate can be 50%.

Life cycles:

1- Jungle (sylvatic) yellow fever: Monkey is the main host and the virus is transmitted between monkeys by **Aedes Africans** mosquito in Africa.

Forest worker/visitor are infected accidentally and carries infection to urban areas.

2- Urban yellow fever:

The disease is introduced into towns; so the cycle is: **man -mosquito-man**.

The mosquito is **Aedes egypti**.

Laboratory diagnosis:

1- Intra-cerebral inoculation of mice with **blood sample**.

2- Direct detection of RNA virus by PCR.

3- Detection of rising antibody titer by neutralizing test, complement fixation, haemagglutination test.

Prevention and control:

1- Use of insecticide to kill mosquitoes and their eggs.

2- Vaccine: "17 D strain" live attenuated virus, given as a single dose subcutaneously, safe long lasting protection is obtained for 10 years.

Vaccination is required for travelers to endemic equatorial zones.

Dengue virus

RNA enveloped viruses.

The natural hosts are monkeys and man, **mosquitoes are the vector.**

Dengue virus has an **urban and a jungle cycle.**

Symptoms include **fever, headache, rash, bone pain ("breakbone fever")**, myalgias and arthralgia.

Dengue Hemorrhagic Fever is a deadly complication of dengue.

Case fatality rate is 50%.

Ebola and Marburg viruses

RNA enveloped viruses cause **fatal disease** in man, there is **haemorrhage**, fever, diarrhea, vomiting and muscle pain.

No known reservoir/ vector hosts.

They are transmitted by direct contact with **infected body fluids.**

MCQS

1- Salmonella spp have the following biochemical reactions **Except:**

- a- Ferment lactose.
- b- Ferment glucose.
- c- Indole negative.
- d- Urease negative.

2- Salmonella causing food poisoning include:

- a- Salmonella typhi.
- b- Salmonella paratyphi A.
- c- Salmonella paratyphi B.
- d- Salmonella typhimurium.

3- Culture of Campylobacter is done under the following condition:

- a- Micro-aerophilic and capnophilic.
- b- Strict aerobe.
- c- Strict anaerobe.
- d- Facultative anaerobe.

4- The laboratory test of choice to diagnose Brucellosis is:

- a- Blood culture.
- b- Gram stain.
- c- Direct culture on nutrient agar.
- d- Blood film.

5- Bacillus cereus is the etiological agent of all of the followings **Except:**

- a- "Short-incubation" or emetic form of food poisoning.
- b- "Long-incubation" or diarrheal form of food poisoning.
- c- Cutaneous Anthrax.
- d- Non bloody diarrhea.

6- Antibiotic therapy may be a critical step in the management of patients infected with the following microorganisms. Which organism listed below would not benefit from prompt antibiotic treatment?

- a- B.anthraxis.
- b- C.difficile.
- c- C.botulinum.
- d- C.perfringens.

7- Rodent are reservoirs of leptospira which colonize their:

- a- Skin.
- b- Kidneys.
- c- Spleen.
- d- Eye.

8- All is true about yellow fever vaccine **Except:**

- a- Killed vaccine.
- b- Safe.
- c- Long lasting protection for 10 years.
- d- Required for travellers to endemic equatorial zones.

Parasitic GI diseases

Blood flukes (Schistosoma)

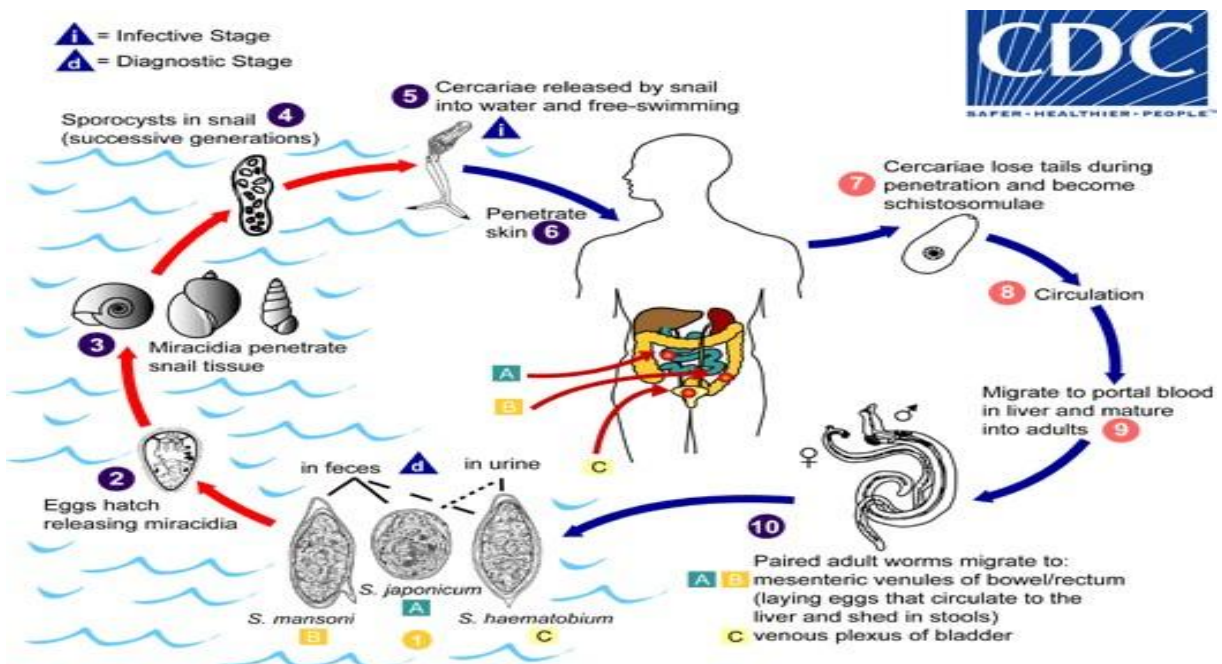
Schistosomes are the causative agents of the disease schistosomiasis.

Three species of schistosomes that parasitize humans are of major importance: *S.mansoni*, *S. haematobium* and *S.japonicum*.

The morphologic characteristics of the three species are given in the following table:

	<i>S.mansoni</i>	<i>S.japonicum</i>
Disease	Intestinal bilharziasis	Intestinal bilharziasis
Distribution	Nile Delta, Africa, South America, Middle East.	Far East.
Intermediate host	<i>Biomphalaria alexandrina</i>	<i>Onchomelania</i> species.
Reservoir host	Monkeys, rodents.	Dogs, cats, rodents, pigs and cattle.
Habitat	Inferior mesenteric plexus of veins	superior mesenteric plexus of veins

Life-cycle:



Eggs escape to the perivascular tissue due to:

- i. pressure within the venule.
- ii. effect of the spine.
- iii. the oozing lytic secretion of the miracidium within the egg. From the perivascular tissue eggs pass to the lumen of the intestine in case of *S.mansoni* and *S.japonicum* and to the lumen of the urinary bladder in case of *S. haematobium*.

When the embryonated eggs (diagnostic stage) discharged in faeces enter fresh water, miracidia hatch from the eggs, rapidly swim by means of cilia (survive for 24 hours).

Then they penetrate the body of snail (intermediate host).

The cercariae infect the definitive host by penetrating the skin, aided by:

- i. proteolytic enzymes secreted from penetration glands.
- ii. the surface tension of drying droplets of water.
- iii. the strong lashing movements of the tail pressing the body of the cercaria into the skin.

Drinking water may lead to infection when cercariae penetrate the mucosa above gastric acidity that kills them.

They lose their tails out and transform into the schistosomula.

- Schistosomula migrate to the lungs via the venous circulation.

They break out from the lungs Through the pulmonary capillaries and are carried through the left heart into the systemic circulation.

They finally reach the intra-hepatic branches of the portal veins in the liver where they settle down and mature.

- Within several weeks, pairing of the worms takes place and they migrate to the mesenteric veins (*S.mansoni* and *S. japonicum*) or to the vesical plexus (*S. haematobium*), where the females lay eggs.

- The prepatent period between penetration by cercariae and the appearance of eggs in the excreta is around 35 days for *S.munsoni* and *Sjaponicum* and about 70 days for *S.haematobium*.

Pathogenesis and clinical picture:

Intestinal schistosomiasis is 7-8 weeks (for *S. mansoni* infection) and 5-6 weeks (for *S. japonicum* infection).

Pathological changes are caused mainly by schistosome eggs deposited in various tissues.

The antigens excreted from embryonated eggs result in formation of granulomas that lead eventually to fibrotic changes in the tissues involved in chronic infections.

Thus, the lesions produced by the infections are essentially the same among the 3 major species of schistosomes. There are 4 progressive stages:

1. Stage of invasion (1 - 4 days):

- Skin lesion due to cercarial penetration.
- Local dermatitis, irritation, papular rash.

2. Stage of migration (3 — 4 weeks):

- Lung: verminous pneumonitis, minute haemorrhages, cough and haemoptysis.
- Liver: enlarged and tender.
- Metabolic products of migrating parasites: result in toxic and allergic manifestations de urticarial, eosinophilia, leukocytosis, fever, and headache and muscle pain.

3. Stage of egg deposition and extrusion (acute stage: 1-2 month):

Eggs deposited in the venous plexus escape from the veins into the perivascular tissue and finally find their way to the outside with urine and stool (open lesion stage).

With extrusion of eggs through the wall of the intestine or bladder, there may be again generalized malaise, fever, rigors, urticaria, abdominal pain and liver tenderness.

- In schistosomiasis mansoni and schistosomiasis japonicum (intestinal bilharziasis):

Active egg deposition results in tissue damage and haemorrhage.

symptoms:

- dysentery with blood and mucus in stool.
- abdominal pain.
- *S. japonicum* may present with Katayama fever.

4. Stage of tissue reaction (chronic stage):

Eggs trapped in the tissue secrete soluble egg antigen that stimulate inflammatory reactions (eosinophilic granulomas) and fibrosis leading to formation of bilharzial nodules, papillomata, sandy patches and reduced egg output (closed - lesion stage).

- In schistosomiasis mansoni and schistosomiasis japonicum:

The intestinal wall becomes fibrosed, thickened and may be complicated with strictures, sinuses, fistula and prolapse.

Eggs that fail to be fixed to venules of intestinal walls fall in the lumen and are swept by the blood to the liver.

This results in periportal fibrosis, portal hypertension, hepatosplenomegaly, followed by shrunken fibrosed liver.

This results in portal hypertension with oesophageal varices, haemorrhoids, and ascites.

This is manifested by hematemesis, melena and distended abdomen.

This stage ends in hepatocellular failure (impaired liver function) and fatal haemorrhage. The enlarged spleen and the hypersplenism lead to destruction of the red blood cells that add the existing anaemia.

Later on it leads to pancytopenia.

II. Intestinal schistosomiasis:

1. Clinical diagnosis: Living in or visiting endemic areas with history of exposure to canal water and complaining of dysentery or hepatosplenomegaly is suggestive.

2. Parasitological diagnosis:

Stool examination for recovery of the eggs (diagnostic stage) by:

a. Direct smear.

b. Concentration techniques may be required when infection is light.

3. Immunodiagnosis:

Detection of antigens and antibodies as in urinary schistosomiasis.

4. Other tests:

a) Complete blood count (CBC): Shows eosinophilia up to 40 % and anaemia.

b) Rectal swab

c) colonoscopy and sigmoidoscopy for visualizing lesions e.g. papillomata, colonic polyps, and for biopsies.

d) Ultrasonography and other imaging techniques: show periportal fibrosis and hepatosplenomegaly.

Treatment:

1. Praziquantel (Biltricide), Oxmequine (Vansil) and Metrifonate

Prevention and control:

1. Mass treatment and follow up of infected persons.

2. Health education

3- Snail control

Taenia saginata

(beef tape worm, bald tape worm)

Geographical distribution

Cosmopolitan especially in cattle raising countries.

Adult lives in the small intestine of man who is the only definitive host attaching itself to the mucosa.

Mode of infection: by ingestion of under-cooked beef containing viable cysticerci usually in the form of hamburgers or grilled meat (Kebab).

In the small intestine, the scolex is evaginated and gets attached to the mucosa and matures in about 3 months.

Habitat: small intestine.

Definitive Host: man.

Intermediate Host: cattle.

Infective stage: cysticercous bovis in cattle meat.

Diagnostic stage: mature egg in stool.

Pathogenesis and clinical picture

1. Intestinal disturbances, hunger pains and diarrhea.
2. Neurologic manifestations.
3. Intestinal obstruction.
4. Loss of weight as the parasite consumes much of the food of the patient leading to malnutrition and avitaminosis.
5. Appendicitis or cholangitis caused by stray segments of the worm.
6. Migrating segments creeping out of the anus cause irritation, itching and worry of the patient

Diagnosis

1. Clinical diagnosis: Suggestive symptoms as diarrhoea and weight loss, with passage of gravid segments through the anus.

2. Parasitological diagnosis:

Stool examination to find eggs and proglottids (diagnostic stages) Recovery of eggs from perianal region using a swab.

Gravid segments in faeces, if not found give a saline purge.

Compress the segments between two slides and count the main lateral branches.

3. Immunodiagnosis: Detection of coproantigen in patients' stool by ELISA.

Treatment

1. Praziquantel is the drug of choice for both *Taenia saginata* and *Taenia solium*; it causes worm paralysis, in a dose of 10mg/Kg as a single oral dose.
2. Yomesan in a single oral dose of 2 gm on empty stomach.

Control

1. Treatment of the infected persons.
2. Avoid grazing cattle on fields fertilized by human manure.
3. Proper meat inspection in slaughter houses.
4. Refrigeration of meat at -10°C for 5 days.
5. Thorough cooking of beef.

Taenia solium (pork tape worm)

Geographical distribution

Cosmopolitan where pork is eaten. Thus it is rare in Islamic and Jewish communities.

Similar to T.saginata but differs in

1. Pig is the intermediate host.
2. Larval stage is the cysticercus cellulosae which differs from cysticercus bovis in having invaginated scolex with four suckers and hooks (infective stage).
3. Man is infected by ingestion of under-cooked pork

Pathogenesis and clinical picture

1- Intestinal disturbances, neurologic manifestations, intestinal obstruction or loss of weight.

2- When man ingests eggs of *T.solium*, the larval stage (cysticercus cellulosae) develops in the extraintestinal tissues leading to a condition known as cysticercosis.

This condition is likely to occur with *T.solium* not with *T.saginata*, a fact which makes *T.solium* runs a more serious course.

Diagnosis

1- Clinical diagnosis.

2- Laboratory diagnosis:

- Examination of stool to detect the gravid segments.
- The eggs are rarely present in stool and are similar to those of *T.saginata*.

Treatment

1. Similar to that of *T.saginata*.

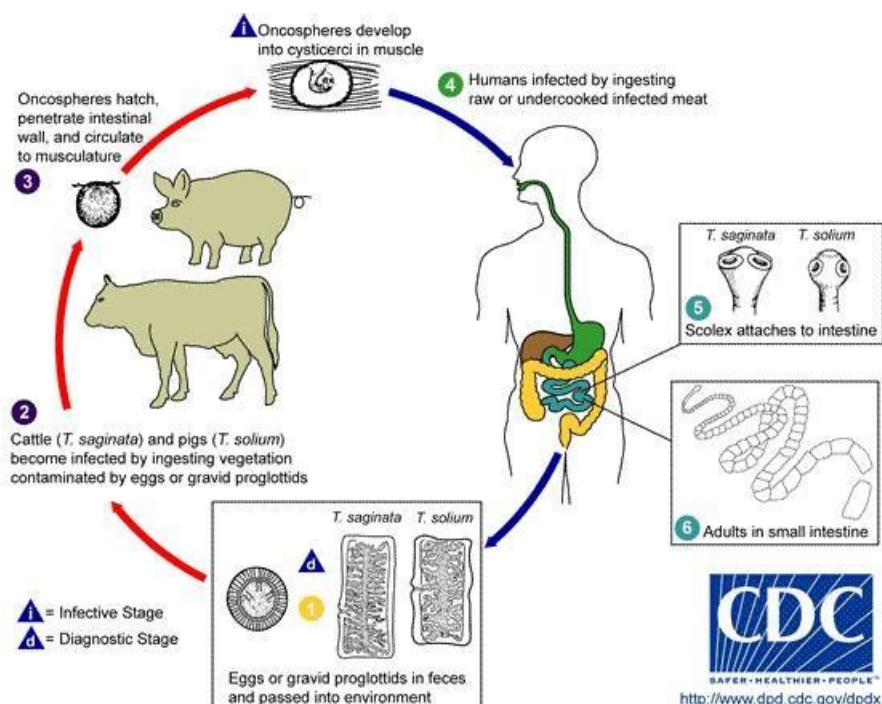
2. Yomesan in a single oral dose of 2gm on empty stomach.

It is recommended to give a saline purge 1-2 hours later to avoid cysticercosis from eggs of disintegrated segments.

Also Quinacrine hydrochloride (atebrine) can be added as it leads to expulsion of the parasite in an intact form.

3. Giving antiemetic before treatment to avoid vomiting.

Life cycle: (Taenia saginata and solium)



Intestinal nematodes

Ascaris lumbricoides

(giant intestinal round worm)

Geographical distribution

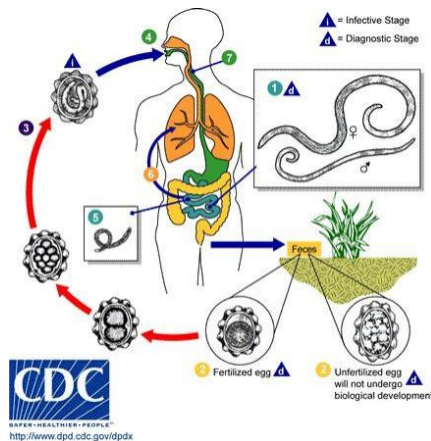
Cosmopolitan. More prevalent in warm countries with bad sanitation. Common in Egypt (especially children).

Host: Man.

Habitat: Small intestine (free in the lumen).

Life cycle: (Geohelminthic infection).

Infection of man takes place by swallowing water or raw vegetables polluted with embryonated eggs containing the infective larva or through contaminated hands by polluted soil.



Pathogenesis

Tissue damage may be due to:

1. Large size of *Ascaris* (largest intestinal nematode).
2. Adults do not attach to intestinal wall, thus they may go to ectopic places.
3. Toxic products stimulate immune response.

Clinical picture

The usual infection (10 - 20 worms) often passes unnoticed.

In heavier infection the following occurs:

1. During larval migration:

There is intense local inflammatory reaction around the larvae leading to pneumonitis manifested by fever, cough, dyspnea and eosinophilia.

In the intestine:

- Abdominal pain, nausea, vomiting, colic, distention or dyspepsia due to production of anti. enzymes that interfere with protein digestion leading to malnutrition and underdevelopment of children.
- Changes of the bowel movements (diarrhoea or constipation).

2. Complications:

- Traumatic effects: due to irritation of adult worms by fever, drugs or anaesthesia.

Intestinal obstruction, volvulus and intussusception occur by a mass of worms.

Worms are not fixed to intestine so may be sloughed to:

- o Bile ducts causing obstructive jaundice.
- o Liver (carrying intestinal bacteria) resulting in liver abscesses.
- o Appendix leading to appendicitis.
- o Ampulla of Vater leading to acute haemorrhagic pancreatitis.
- o Peritoneum leading to peritonitis.
- o Stomach leading to vomiting or escape through external nares.
- o Very rarely to the trachea leading to suffocation.

- Toxic effects:

The by-products of living and dead worms may produce toxic symptoms in sensitized persons such as oedema of the face, urticaria, asthma, insomnia, nervous irritability and convulsions.

Diagnosis

1- Clinical diagnosis: non specific.

2- Parasitological diagnosis:

Detection of eggs in the stool.

Detection of larvae in the sputum and Charcot Leyden crystals early in the disease.

3- other tests:

- High eosinophilia.
- X-ray on abdomen after barium meal reveals worms as filling defects

Treatment

Mebendazole or flubendazole: 100 mg twice per day for 3 days.

Prevention and control

- 1- Purification and chlorination of water.
- 2- Proper washing of raw vegetables and washing hands before meals.
- 3- Proper disposal of sewage (human faeces).
- 4- Mass treatment, personal hygiene and education of people.

Enterobius vermicularis

(Pin worm, Seat worm)

Geographical distribution

Cosmopolitan, more in temperate and cold climates.

Final host: Man especially children.

Habitat: Large intestine mainly caecum, colon, rectum and appendix.

Mode of infection

1- Ingestion of infective eggs through:

- Auto-infection (contaminated hand to mouth).
- Contaminated food or drinks.

2- Air-borne infection through inhalation of eggs in dust particles.

Pathogenesis and symptoms

- 1) Irritation, itching and scratching of the peri-anal skin (pruritis ani), lead to insomnia, nervous manifestations and secondary bacterial infection.
- 2) Worms in young girls may deposit eggs on vulva producing pruritis vulvae.
- 3) Worms may migrate to the vagina (vaginitis), uterus, urinary bladder (urethritis) leading to irritation and inflammation.
- 4) Appendicitis is due to migrating adults.
- 5) Due to the easy spread of infection from one member of the family to another, all the family should be examined and all infected ones should be treated at the same time to get rid of the infection among the whole family.
- 6) Sometimes, the vagina and bladder are invaded in female patients causing irritation.
- 7) Migration to the peritoneal cavity (through uterine tubes) will cause pelvic peritonitis.

Diagnosis

1- Clinically: Infection is suspected in children with Pruritus ani at night.

2- Laboratory:

- Adult worms may be seen in stool or anal area.
- Eggs are rarely found in stool (about 5 % only).
- Swabbing of anal or perianal area by:
 - a) N.I.H. swab (National Institute of Health)
 - b) Scotch adhesive tape swab

Treatment

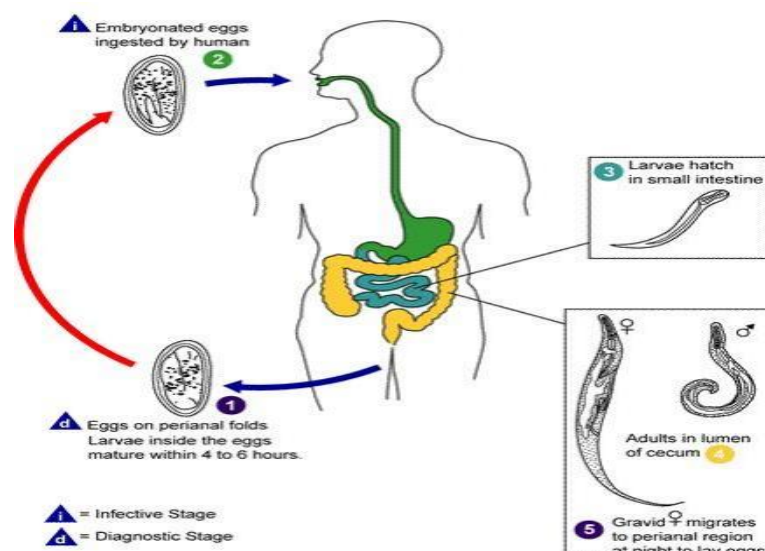
1. Albendazole, Mebendazole.

2. Local application of white oxide of mercury (white precipitate ointment) around the anus at night (relieves itching, kills female worms coming out and prevents dispersal of eggs).

Prevention and control

- Personal hygiene.
- Infected persons should sleep alone, their underwear and bedding carefully handled & washed.
- Food and drink should be protected from dust & hand.

Life cycle



Extra-intestinal amoebiasis

Hepatic amoebiasis

Pathogenesis

Hepatic amoebiasis may occur due to either:

- a. Direct spread** of amoebic trophozoites from a primary intestinal lesion to surrounding tissues.
- b. Hematogenous spread** through invasion of blood , Portal circulation draining the intestine carries amoebic emboli mainly to the right lobe of the liver.

Hepatic amoebiasis: liver affection may be in the form of diffuse amoebic hepatitis or liver abscess.

a. Diffuse Amoebic Hepatitis:

In considerable number of patients with amoebic colitis there is an associated enlargement of the liver.

This condition starts with constant invasion of the liver by amoebae from an active intestinal process via the portal circulation.

b. Amoebic liver Abscess:

In foci where amoebic colonization continues, there is lytic necrosis of hepatic parenchyma.

The lesions coalesce forming typically a solitary lesion.

It begins as a small area of necrosed hepatic tissue, which increases in size with liquefaction in the centre and thickening of the wall.

Its contents are lysed liver cells, bile,fat,RBCs and collagen strands; all forming a viscid chocolate red or creamy mass (anchovy-sauce pus).

Amoebic trophozoites are only found in the edge of the lesion where they multiply.

The commonest site is the posterior part of the right lobe (silent area) causing upward bulging of the diaphragm.

The hepatic abscess may become chronic, absorbed, fibrosed, calcified or may extend to other organs.

Clinical picture

a. Symptoms:

There is usually an acute onset of pain in the upper abdomen with high fever, the pain is intense and constant, radiating to the scapular region and right shoulder; it increases with deep breathing or coughing.

Fever is varying from 38° and 40° C.

There is anorexia, nausea, vomiting and fatigue.

b. Signs:

1. Enlarged tender liver
2. Pallor
3. Earthy look
4. Fever
5. Tinge of jaundice

Differential diagnosis

Other causes of hepatomegaly and cystic lesions in the liver (hydatid disease).

Diagnosis

a. Clinical diagnosis: clinical picture

b. Laboratory diagnosis:

1. Indirect by serological tests: to detect specific antibodies.

The indirect haemagglutination test (IHAT) and ELISA technique and both sensitive and specific but cannot be used to assess cure because antibodies persist for a long time.

2. Direct diagnosis by:

- i. Aspiration of the abscess: specimen is taken from the border of the abscess for detection of trophozoite.
- ii. Stool examination: may be positive in only one third of cases.
- iii. Detection of copro-antigen (in stool) by enzyme immunoassay.

c. Radiology:

- i. X-ray shows high density in hepatic shadow and abnormal elevation of the right dome of diaphragm
- ii. Ultrasonography, computed tomography (CT) and magnetic resonance may visualize the amoebic liver abscess.

d. Complete blood count (CBC) shows polymorphonuclear leucocytosis in Amoebic Liver Abscess (differentiated from viral hepatitis).

Prevention and control

1. Personal hygiene
2. Washing of fresh-eaten vegetables and fruits.
3. Safe water supply, water from polluted source should be boiled.
4. Insect control or protection of food from insects.
5. Treatment of patients and examination of contacts.